

Microsoft Windows [Version 10.0.26200.8655]

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C:\Users\Y.Lekhasree>mysql -u root -p

Enter password: *****

Welcome to the MySQL monitor. Commands end with ; or \g.

Your MySQL connection id is 20

Server version: 8.0.46 MySQL Community Server - GPL

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Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> create database PMS;

Query OK, 1 row affected (0.12 sec)

mysql> use PMS;

Database changed

mysql> CREATE TABLE departments (

-> dept_id INT PRIMARY KEY,

-> dept_name VARCHAR(50) UNIQUE NOT NULL);

Query OK, 0 rows affected (0.14 sec)

mysql> INSERT INTO departments (dept_id, dept_name) VALUES

-> (101, 'CSE'),

-> (102, 'ECE'),

-> (103, 'EEE'),

-> (104, 'Mechanical'),

-> (105, 'Civil');

Query OK, 5 rows affected (0.02 sec)

Records: 5 Duplicates: 0 Warnings: 0

```
mysql> SELECT * FROM departments;
```

```
+-----+-----+
| dept_id | dept_name |
+-----+-----+
| 105 | Civil |
| 101 | CSE |
| 102 | ECE |
| 103 | EEE |
| 104 | Mechanical |
+-----+-----+
```

5 rows in set (0.00 sec)

```
mysql> CREATE TABLE students (
```

```
-> student_id INT AUTO_INCREMENT PRIMARY KEY,
-> student_name VARCHAR(100) NOT NULL,
-> email VARCHAR(100) UNIQUE,
-> phone BIGINT,
-> cgpa DECIMAL(3,1),
-> dept_id INT,
-> CONSTRAINT fk_student_department
-> FOREIGN KEY (dept_id)
-> REFERENCES departments(dept_id) );
```

Query OK, 0 rows affected (0.07 sec)

```
mysql> INSERT INTO students
```

```
-> (student_name, email, phone, cgpa, dept_id)
-> VALUES
-> ('Saran','saran@gmail.com',9876543210,8.5,101),
-> ('Priya','priya@gmail.com',9876543211,9.0,102),
```

```
-> ('Rahul','rahul@gmail.com',9876543212,7.8,101),
-> ('Deepa','deepa@gmail.com',9876543213,8.9,103),
-> ('Anjali','anjali@gmail.com',9876543214,8.2,102),
-> ('Kumar','kumar@gmail.com',9876543215,7.5,104),
-> ('Sneha','sneha@gmail.com',9876543216,9.1,101),
-> ('Varun','varun@gmail.com',9876543217,8.0,105),
-> ('Charan','charan@gmail.com',9876543218,7.2,104),
-> ('Harsha','harsha@gmail.com',9876543219,8.7,101);
```

Query OK, 10 rows affected (0.05 sec)

Records: 10 Duplicates: 0 Warnings: 0

```
mysql> SELECT * FROM students;
```

```
+-----+-----+-----+-----+-----+
| student_id | student_name | email      | phone   | cgpa | dept_id |
+-----+-----+-----+-----+-----+
| 1 | Saran    | saran@gmail.com | 9876543210 | 8.5 | 101 |
| 2 | Priya    | priya@gmail.com | 9876543211 | 9.0 | 102 |
| 3 | Rahul    | rahul@gmail.com | 9876543212 | 7.8 | 101 |
| 4 | Deepa    | deepa@gmail.com | 9876543213 | 8.9 | 103 |
| 5 | Anjali    | anjali@gmail.com | 9876543214 | 8.2 | 102 |
| 6 | Kumar    | kumar@gmail.com | 9876543215 | 7.5 | 104 |
| 7 | Sneha    | sneha@gmail.com | 9876543216 | 9.1 | 101 |
| 8 | Varun    | varun@gmail.com | 9876543217 | 8.0 | 105 |
| 9 | Charan    | charan@gmail.com | 9876543218 | 7.2 | 104 |
| 10 | Harsha    | harsha@gmail.com | 9876543219 | 8.7 | 101 |
+-----+-----+-----+-----+-----+
```

10 rows in set (0.00 sec)

```
mysql> CREATE TABLE companies (
```

```
-> company_id INT PRIMARY KEY,
```

```
-> company_name VARCHAR(100) UNIQUE,  
-> location VARCHAR(100),  
-> package_lpa DECIMAL(4,1) );
```

Query OK, 0 rows affected (0.10 sec)

```
mysql> INSERT INTO companies (company_id, company_name, location, package_lpa)
```

```
-> VALUES
```

```
-> (1, 'TCS', 'Hyderabad', 4.5),  
-> (2, 'Infosys', 'Bangalore', 5.2),  
-> (3, 'Accenture', 'Chennai', 6.0),  
-> (4, 'Wipro', 'Pune', 4.2),  
-> (5, 'Cognizant', 'Hyderabad', 5.8);
```

Query OK, 5 rows affected (0.05 sec)

Records: 5 Duplicates: 0 Warnings: 0

```
mysql> SELECT * FROM companies;
```

```
+-----+-----+-----+-----+  
| company_id | company_name | location | package_lpa |  
+-----+-----+-----+-----+  
| 1 | TCS | Hyderabad | 4.5 |  
| 2 | Infosys | Bangalore | 5.2 |  
| 3 | Accenture | Chennai | 6.0 |  
| 4 | Wipro | Pune | 4.2 |  
| 5 | Cognizant | Hyderabad | 5.8 |  
+-----+-----+-----+-----+
```

5 rows in set (0.00 sec)

```
mysql> CREATE TABLE jobs (
```

```
-> job_id INT PRIMARY KEY,  
-> company_id INT,  
-> role_name VARCHAR(100),
```

```
-> min_cgpa DECIMAL(3,1),
-> CONSTRAINT fk_job_company
-> FOREIGN KEY (company_id)
-> REFERENCES companies(company_id) );
```

Query OK, 0 rows affected (0.06 sec)

```
mysql> INSERT INTO jobs (job_id, company_id, role_name, min_cgpa) VALUES
```

```
-> (1, 1, 'Java Developer', 7.5),
-> (2, 2, 'Python Developer', 8.0),
-> (3, 3, 'Full Stack Developer', 8.2),
-> (4, 4, 'Data Analyst', 7.0),
-> (5, 5, 'Software Engineer', 7.8);
```

Query OK, 5 rows affected (0.05 sec)

Records: 5 Duplicates: 0 Warnings: 0

```
mysql> SELECT * FROM jobs;
```

```
+-----+-----+-----+-----+
| job_id | company_id | role_name      | min_cgpa |
+-----+-----+-----+-----+
| 1 | 1 | Java Developer | 7.5 |
| 2 | 2 | Python Developer | 8.0 |
| 3 | 3 | Full Stack Developer | 8.2 |
| 4 | 4 | Data Analyst | 7.0 |
| 5 | 5 | Software Engineer | 7.8 |
+-----+-----+-----+-----+
```

5 rows in set (0.00 sec)

```
mysql> CREATE TABLE applications (
```

```
-> application_id INT PRIMARY KEY,
-> student_id INT,
-> job_id INT,
```

```

-> status VARCHAR(20),
-> CONSTRAINT fk_application_student
-> FOREIGN KEY (student_id)
-> REFERENCES students(student_id),
-> CONSTRAINT fk_application_job
-> FOREIGN KEY (job_id)
-> REFERENCES jobs(job_id) );

```

Query OK, 0 rows affected (0.15 sec)

mysql> INSERT INTO applications

```

-> (application_id, student_id, job_id, status) VALUES
-> (1, 1, 1, 'Applied'),
-> (2, 2, 2, 'Selected'),
-> (3, 3, 1, 'Rejected'),
-> (4, 4, 3, 'Applied'),
-> (5, 5, 2, 'Selected'),
-> (6, 6, 4, 'Rejected'),
-> (7, 7, 5, 'Selected'),
-> (8, 8, 3, 'Applied'),
-> (9, 9, 4, 'Rejected'),
-> (10, 10, 5, 'Selected');

```

Query OK, 10 rows affected (0.04 sec)

Records: 10 Duplicates: 0 Warnings: 0

mysql> SELECT * FROM applications;

```

+-----+-----+-----+-----+
| application_id | student_id | job_id | status |
+-----+-----+-----+-----+
|          1 |          1 |          1 | Applied |
|          2 |          2 |          2 | Selected |

```

	3		3		1		Rejected	
	4		4		3		Applied	
	5		5		2		Selected	
	6		6		4		Rejected	
	7		7		5		Selected	
	8		8		3		Applied	
	9		9		4		Rejected	
	10		10		5		Selected	

```

+-----+-----+-----+-----+

```

10 rows in set (0.00 sec)

```

mysql> CREATE TABLE aptitude_tests (
-> test_id INT PRIMARY KEY,
-> student_id INT,
-> marks INT,
-> CONSTRAINT fk_aptitude_student
-> FOREIGN KEY (student_id)
-> REFERENCES students(student_id) );

```

Query OK, 0 rows affected (0.12 sec)

```

mysql> INSERT INTO aptitude_tests (test_id, student_id, marks)
-> VALUES
-> (1, 1, 85),
-> (2, 2, 92),
-> (3, 3, 78),
-> (4, 4, 88),
-> (5, 5, 81),
-> (6, 6, 75),
-> (7, 7, 95),
-> (8, 8, 80),

```

-> (9, 9, 72),

-> (10, 10, 89);

Query OK, 10 rows affected (0.04 sec)

Records: 10 Duplicates: 0 Warnings: 0

mysql> SELECT * FROM aptitude_tests;

+-----+-----+-----+		
test_id student_id marks		
+-----+-----+-----+		
1	1	85
2	2	92
3	3	78
4	4	88
5	5	81
6	6	75
7	7	95
8	8	80
9	9	72
10	10	89
+-----+-----+-----+		

10 rows in set (0.00 sec)

mysql> CREATE TABLE interviews (

-> interview_id INT PRIMARY KEY,

-> student_id INT,

-> company_id INT,

-> round_name VARCHAR(50),

-> result VARCHAR(20),

-> CONSTRAINT fk_interview_student

-> FOREIGN KEY (student_id)


```
-> REFERENCES students(student_id),
-> CONSTRAINT fk_interview_company
-> FOREIGN KEY (company_id)
-> REFERENCES companies(company_id) );
```

Query OK, 0 rows affected (0.13 sec)

```
mysql> INSERT INTO interviews
```

```
-> (interview_id, student_id, company_id, round_name, result)
-> VALUES
-> (1, 1, 1, 'Technical Round', 'Passed'),
-> (2, 2, 2, 'HR Round', 'Passed'),
-> (3, 3, 1, 'Technical Round', 'Failed'),
-> (4, 4, 3, 'Managerial Round', 'Passed'),
-> (5, 5, 2, 'HR Round', 'Passed'),
-> (6, 6, 4, 'Technical Round', 'Failed'),
-> (7, 7, 5, 'Technical Round', 'Passed'),
-> (8, 8, 3, 'HR Round', 'Passed'),
-> (9, 9, 4, 'Technical Round', 'Failed'),
-> (10, 10, 5, 'Managerial Round', 'Passed');
```

Query OK, 10 rows affected (0.05 sec)

Records: 10 Duplicates: 0 Warnings: 0

```
mysql> SELECT * FROM interviews;
```

```
+-----+-----+-----+-----+-----+
| interview_id | student_id | company_id | round_name | result |
+-----+-----+-----+-----+-----+
| 1 | 1 | 1 | Technical Round | Passed |
| 2 | 2 | 2 | HR Round | Passed |
```

	3		3		1		Technical Round		Failed	
	4		4		3		Managerial Round		Passed	
	5		5		2		HR Round		Passed	
	6		6		4		Technical Round		Failed	
	7		7		5		Technical Round		Passed	
	8		8		3		HR Round		Passed	
	9		9		4		Technical Round		Failed	
	10		10		5		Managerial Round		Passed	

```

+-----+-----+-----+-----+-----+

```

10 rows in set (0.00 sec)

```

mysql> CREATE TABLE placements (
->  placement_id INT PRIMARY KEY,
->  student_id INT,
->  company_id INT,
->  package_lpa DECIMAL(4,1),
->
->  CONSTRAINT fk_placement_student
->  FOREIGN KEY (student_id)
->  REFERENCES students(student_id),
->
->  CONSTRAINT fk_placement_company
->  FOREIGN KEY (company_id)
->  REFERENCES companies(company_id)
-> );

```

Query OK, 0 rows affected (0.12 sec)

```

mysql> INSERT INTO placements

```

-> (placement_id, student_id, company_id, package_lpa)

-> VALUES

-> (1, 2, 2, 5.2),

-> (2, 5, 2, 5.2),

-> (3, 7, 5, 5.8),

-> (4, 10, 5, 5.8),

-> (5, 4, 3, 6.0);

Query OK, 5 rows affected (0.04 sec)

Records: 5 Duplicates: 0 Warnings: 0

mysql> SELECT * FROM placements;

placement_id	student_id	company_id	package_lpa
1	2	2	5.2
2	5	2	5.2
3	7	5	5.8
4	10	5	5.8
5	4	3	6.0

5 rows in set (0.00 sec)

mysql> SELECT

-> s.student_name,

-> d.dept_name

-> FROM students s

-> INNER JOIN departments d

-> ON s.dept_id = d.dept_id;

```
+-----+-----+
| student_name | dept_name |
```

```
+-----+-----+
```

```
| Varun   | Civil   |
| Saran   | CSE     |
| Rahul   | CSE     |
| Sneha   | CSE     |
| Harsha  | CSE     |
| Priya   | ECE     |
| Anjali  | ECE     |
| Deepa   | EEE     |
| Kumar   | Mechanical |
| Charan  | Mechanical |
```

```
+-----+-----+
```

10 rows in set (0.00 sec)

mysql> SELECT

-> s.student_name,

-> c.company_name

-> FROM applications a

-> JOIN students s ON a.student_id = s.student_id

-> JOIN jobs j ON a.job_id = j.job_id

-> JOIN companies c ON j.company_id = c.company_id;

```
+-----+-----+
```

```
| student_name | company_name |
```

```
+-----+-----+
```

```
| Saran   | TCS     |
| Rahul   | TCS     |
```

Priya	Infosys	
Anjali	Infosys	
Deepa	Accenture	
Varun	Accenture	
Kumar	Wipro	
Charan	Wipro	
Sneha	Cognizant	
Harsha	Cognizant	

+-----+-----+

10 rows in set (0.04 sec)

mysql> SELECT

-> student_name,

-> cgpa

-> FROM students

-> WHERE cgpa > 8;

+-----+-----+

student_name	cgpa	
--------------	------	--

+-----+-----+

Saran	8.5	
-------	-----	--

Priya	9.0	
-------	-----	--

Deepa	8.9	
-------	-----	--

Anjali	8.2	
--------	-----	--

Sneha	9.1	
-------	-----	--

Harsha	8.7	
--------	-----	--

+-----+-----+

6 rows in set (0.00 sec)

```
mysql> SELECT *
```

```
-> FROM companies
```

```
-> WHERE package_lpa > 5;
```

```
+-----+-----+-----+-----+
| company_id | company_name | location | package_lpa |
+-----+-----+-----+-----+
|      2 | Infosys    | Bangalore |      5.2 |
|      3 | Accenture  | Chennai  |      6.0 |
|      5 | Cognizant  | Hyderabad |      5.8 |
+-----+-----+-----+-----+
```

```
3 rows in set (0.00 sec)
```

```
mysql> SELECT DISTINCT
```

```
-> s.student_id,
```

```
-> s.student_name
```

```
-> FROM students s
```

```
-> JOIN interviews i
```

```
-> ON s.student_id = i.student_id;
```

```
+-----+-----+
| student_id | student_name |
+-----+-----+
|      1 | Saran      |
|      2 | Priya      |
|      3 | Rahul      |
|      4 | Deepa      |
|      5 | Anjali     |
|      6 | Kumar      |
|      7 | Sneha      |
```

```
|      8 | Varun      |
|      9 | Charan     |
|     10 | Harsha     |
```

```
+-----+-----+
```

10 rows in set (0.04 sec)

mysql> SELECT

-> s.student_id,

-> s.student_name

-> FROM students s

-> JOIN placements p

-> ON s.student_id = p.student_id;

```
+-----+-----+
```

```
| student_id | student_name |
```

```
+-----+-----+
```

```
|      2 | Priya      |
```

```
|      4 | Deepa      |
```

```
|      5 | Anjali     |
```

```
|      7 | Sneha      |
```

```
|     10 | Harsha     |
```

```
+-----+-----+
```

5 rows in set (0.00 sec)

mysql> SELECT

-> student_id,

-> student_name

-> FROM students

-> WHERE student_id NOT IN (

```
-> SELECT student_id
```

```
-> FROM placements
```

```
-> );
```

```
+-----+-----+  
| student_id | student_name |
```

```
+-----+-----+
```

```
| 1 | Saran |
```

```
| 3 | Rahul |
```

```
| 6 | Kumar |
```

```
| 8 | Varun |
```

```
| 9 | Charan |
```

```
+-----+-----+
```

```
5 rows in set (0.04 sec)
```

```
mysql> SELECT COUNT(*) AS total_students
```

```
-> FROM students;
```

```
+-----+
```

```
| total_students |
```

```
+-----+
```

```
| 10 |
```

```
+-----+
```

```
1 row in set (0.00 sec)
```

```
mysql> SELECT AVG(cgpa) AS average_cgpa
```

```
-> FROM students;
```

```
+-----+
```

```
| average_cgpa |
```

```
+-----+
```



```
| 8.29000 |
```

```
+-----+
```

```
1 row in set (0.01 sec)
```

```
mysql> SELECT MAX(cgpa) AS highest_cgpa
```

```
-> FROM students;
```

```
+-----+
```

```
| highest_cgpa |
```

```
+-----+
```

```
| 9.1 |
```

```
+-----+
```

```
1 row in set (0.00 sec)
```

```
mysql> SELECT MIN(cgpa) AS lowest_cgpa
```

```
-> FROM students;
```

```
+-----+
```

```
| lowest_cgpa |
```

```
+-----+
```

```
| 7.2 |
```

```
+-----+
```

```
1 row in set (0.00 sec)
```

```
mysql> SELECT
```

```
-> d.dept_name,
```

```
-> COUNT(s.student_id) AS student_count
```

```
-> FROM departments d
```

```
-> JOIN students s
```

```
-> ON d.dept_id = s.dept_id
```

-> GROUP BY d.dept_name;

```
+-----+-----+
| dept_name | student_count |
+-----+-----+
| Civil    |      1 |
| CSE      |      4 |
| ECE      |      2 |
| EEE      |      1 |
| Mechanical |      2 |
+-----+-----+
```

5 rows in set (0.00 sec)

mysql> SELECT

-> c.company_name,

-> COUNT(p.student_id) AS placed_student_count

-> FROM companies c

-> JOIN placements p

-> ON c.company_id = p.company_id

-> GROUP BY c.company_name;

```
+-----+-----+
| company_name | placed_student_count |
+-----+-----+
| Accenture    |      1 |
| Cognizant    |      2 |
| Infosys     |      2 |
+-----+-----+
```

3 rows in set (0.00 sec)

```
mysql> SELECT
```

```
-> d.dept_name,  
-> COUNT(s.student_id) AS student_count  
-> FROM departments d  
-> JOIN students s  
-> ON d.dept_id = s.dept_id  
-> GROUP BY d.dept_name  
-> HAVING COUNT(s.student_id) > 2;
```

```
+-----+-----+
```

```
| dept_name | student_count |
```

```
+-----+-----+
```

```
| CSE      |          4 |
```

```
+-----+-----+
```

```
1 row in set (0.00 sec)
```

```
mysql> SELECT
```

```
-> c.company_name,  
-> COUNT(p.student_id) AS selected_students  
-> FROM companies c  
-> JOIN placements p  
-> ON c.company_id = p.company_id  
-> GROUP BY c.company_name  
-> HAVING COUNT(p.student_id) > 1;
```

```
+-----+-----+
```

```
| company_name | selected_students |
```

```
+-----+-----+
```

```
| Cognizant   |          2 |
```

```
| Infosys    |          2 |
```

```
+-----+-----+
```

2 rows in set (0.00 sec)

```
mysql> SELECT
```

```
-> d.dept_name,
```

```
-> ROUND(AVG(s.cgpa), 2) AS avg_cgpa
```

```
-> FROM departments d
```

```
-> JOIN students s
```

```
-> ON d.dept_id = s.dept_id
```

```
-> GROUP BY d.dept_name
```

```
-> HAVING AVG(s.cgpa) > 8;
```

```
+-----+-----+
```

```
| dept_name | avg_cgpa |
```

```
+-----+-----+
```

```
| CSE      | 8.53 |
```

```
| ECE      | 8.60 |
```

```
| EEE      | 8.90 |
```

```
+-----+-----+
```

3 rows in set (0.00 sec)

```
mysql> SELECT
```

```
-> s.student_name,
```

```
-> d.dept_name
```

```
-> FROM students s
```

```
-> INNER JOIN departments d
```

```
-> ON s.dept_id = d.dept_id;
```

```
+-----+-----+
```

```
| student_name | dept_name |
```

```

+-----+-----+
| Varun   | Civil   |
| Saran   | CSE     |
| Rahul   | CSE     |
| Sneha   | CSE     |
| Harsha  | CSE     |
| Priya   | ECE     |
| Anjali  | ECE     |
| Deepa   | EEE     |
| Kumar   | Mechanical |
| Charan  | Mechanical |
+-----+-----+
10 rows in set (0.00 sec)

```

```

mysql> SELECT
-> s.student_name,
-> c.company_name
-> FROM applications a
-> INNER JOIN students s
-> ON a.student_id = s.student_id
-> INNER JOIN jobs j
-> ON a.job_id = j.job_id
-> INNER JOIN companies c
-> ON j.company_id = c.company_id;
+-----+-----+
| student_name | company_name |
+-----+-----+
| Saran        | TCS          |

```

Rahul	TCS	
Priya	Infosys	
Anjali	Infosys	
Deepa	Accenture	
Varun	Accenture	
Kumar	Wipro	
Charan	Wipro	
Sneha	Cognizant	
Harsha	Cognizant	

+-----+-----+

10 rows in set (0.00 sec)

mysql> SELECT

```
-> c.company_name,
-> j.role_name
-> FROM companies c
-> INNER JOIN jobs j
-> ON c.company_id = j.company_id;
```

+-----+-----+

company_name	role_name	
--------------	-----------	--

+-----+-----+

Accenture	Full Stack Developer	
-----------	----------------------	--

Cognizant	Software Engineer	
-----------	-------------------	--

Infosys	Python Developer	
---------	------------------	--

TCS	Java Developer	
-----	----------------	--

Wipro	Data Analyst	
-------	--------------	--

+-----+-----+

5 rows in set (0.00 sec)

```
mysql> 43
```

```
-> ;
```

ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near '43' at line 1

```
mysql> SELECT
```

```
-> s.student_name,
```

```
-> j.role_name,
```

```
-> a.status
```

```
-> FROM students s
```

```
-> JOIN applications a
```

```
-> ON s.student_id = a.student_id
```

```
-> JOIN jobs j
```

```
-> ON a.job_id = j.job_id;
```

```
+-----+-----+-----+
| student_name | role_name      | status |
+-----+-----+-----+
| Saran        | Java Developer | Applied |
| Rahul        | Java Developer | Rejected |
| Priya        | Python Developer | Selected |
| Anjali       | Python Developer | Selected |
| Deepa        | Full Stack Developer | Applied |
| Varun        | Full Stack Developer | Applied |
| Kumar        | Data Analyst   | Rejected |
| Charan       | Data Analyst   | Rejected |
| Sneha        | Software Engineer | Selected |
| Harsha       | Software Engineer | Selected |
+-----+-----+-----+
```

10 rows in set (0.00 sec)

```
mysql> SELECT
```

```
-> s.student_id,  
-> s.student_name  
-> FROM students s  
-> LEFT JOIN applications a  
-> ON s.student_id = a.student_id  
-> WHERE a.application_id IS NULL;
```

Empty set (0.00 sec)

```
mysql> SELECT *
```

```
-> FROM students  
-> WHERE cgpa = (  
-> SELECT MAX(cgpa)  
-> FROM students  
-> );
```

```
+-----+-----+-----+-----+-----+  
| student_id | student_name | email      | phone   | cgpa | dept_id |  
+-----+-----+-----+-----+-----+  
|      7 | Sneha      | sneha@gmail.com | 9876543216 | 9.1 | 101 |  
+-----+-----+-----+-----+-----+
```

1 row in set (0.00 sec)

```
mysql> SELECT *
```

```
-> FROM students  
-> WHERE cgpa > (  
-> SELECT AVG(cgpa)
```


-> FROM students

->);

```
+-----+-----+-----+-----+-----+
| student_id | student_name | email      | phone   | cgpa | dept_id |
+-----+-----+-----+-----+-----+
| 1 | Saran    | saran@gmail.com | 9876543210 | 8.5 | 101 |
| 2 | Priya    | priya@gmail.com | 9876543211 | 9.0 | 102 |
| 4 | Deepa    | deepa@gmail.com | 9876543213 | 8.9 | 103 |
| 7 | Sneha    | sneha@gmail.com | 9876543216 | 9.1 | 101 |
| 10 | Harsha   | harsha@gmail.com | 9876543219 | 8.7 | 101 |
+-----+-----+-----+-----+-----+
```

5 rows in set (0.00 sec)

mysql> SELECT *

-> FROM companies

-> WHERE package_lpa = (

-> SELECT MAX(package_lpa)

-> FROM companies

->);

```
+-----+-----+-----+-----+
| company_id | company_name | location | package_lpa |
+-----+-----+-----+-----+
| 3 | Accenture | Chennai | 6.0 |
+-----+-----+-----+-----+
```

1 row in set (0.00 sec)

mysql> SELECT *

-> FROM companies

```
-> WHERE package_lpa = (
-> SELECT MAX(package_lpa)
-> FROM companies
-> );
```

```
+-----+-----+-----+-----+
| company_id | company_name | location | package_lpa |
+-----+-----+-----+-----+
|      3 | Accenture   | Chennai |      6.0 |
+-----+-----+-----+-----+
```

1 row in set (0.00 sec)

mysql> SELECT

```
-> s.student_name,
-> a.status
-> FROM students s
-> JOIN applications a
-> ON s.student_id = a.student_id
-> WHERE a.status = 'Rejected';
```

```
+-----+-----+
| student_name | status |
+-----+-----+
| Rahul       | Rejected |
| Kumar       | Rejected |
| Charan      | Rejected |
+-----+-----+
```

3 rows in set (0.00 sec)

mysql> CREATE VIEW selected_students AS

```

-> SELECT
->   s.student_name,
->   c.company_name,
->   p.package_lpa
-> FROM placements p
-> JOIN students s
->   ON p.student_id = s.student_id
-> JOIN companies c
->   ON p.company_id = c.company_id;

```

Query OK, 0 rows affected (0.05 sec)

```
mysql> SELECT * FROM selected_students;
```

```

+-----+-----+-----+
| student_name | company_name | package_lpa |
+-----+-----+-----+
| Priya       | Infosys     | 5.2 |
| Anjali      | Infosys     | 5.2 |
| Sneha       | Cognizant    | 5.8 |
| Harsha      | Cognizant    | 5.8 |
| Deepa       | Accenture    | 6.0 |
+-----+-----+-----+

```

5 rows in set (0.00 sec)

```
mysql> CREATE VIEW high_cgpa_students AS
```

```

-> SELECT
->   student_id,
->   student_name,
->   cgpa
-> FROM students
-> WHERE cgpa > 8;

```

Query OK, 0 rows affected (0.01 sec)

```
mysql> DELIMITER //
```

```
mysql> CREATE PROCEDURE DisplayAllStudents()
```

```
-> BEGIN
```

```
-> SELECT * FROM students;
```

```
-> END //
```

Query OK, 0 rows affected (0.02 sec)

```
mysql> DELIMITER ;
```

```
mysql> CALL DisplayAllStudents();
```

```
+-----+-----+-----+-----+-----+
| student_id | student_name | email      | phone   | cgpa | dept_id |
+-----+-----+-----+-----+-----+
| 1 | Saran    | saran@gmail.com | 9876543210 | 8.5 | 101 |
| 2 | Priya    | priya@gmail.com | 9876543211 | 9.0 | 102 |
| 3 | Rahul    | rahul@gmail.com | 9876543212 | 7.8 | 101 |
| 4 | Deepa    | deepa@gmail.com | 9876543213 | 8.9 | 103 |
| 5 | Anjali   | anjali@gmail.com | 9876543214 | 8.2 | 102 |
| 6 | Kumar    | kumar@gmail.com | 9876543215 | 7.5 | 104 |
| 7 | Sneha    | sneha@gmail.com | 9876543216 | 9.1 | 101 |
| 8 | Varun    | varun@gmail.com | 9876543217 | 8.0 | 105 |
| 9 | Charan   | charan@gmail.com | 9876543218 | 7.2 | 104 |
| 10 | Harsha   | harsha@gmail.com | 9876543219 | 8.7 | 101 |
+-----+-----+-----+-----+-----+
```

10 rows in set (0.00 sec)

Query OK, 0 rows affected (0.07 sec)

```
mysql> DELIMITER //
```

```
mysql> CREATE PROCEDURE GetStudentsByDepartment(IN deptId INT)
```

```
-> BEGIN
```

```
-> SELECT * FROM students WHERE dept_id = deptId;
```

```
-> END //
```

Query OK, 0 rows affected (0.02 sec)

```
mysql> DELIMITER ;
```

```
mysql> CALL GetStudentsByDepartment(101);
```

```
+-----+-----+-----+-----+-----+
| student_id | student_name | email      | phone  | cgpa | dept_id |
+-----+-----+-----+-----+-----+
| 1 | Saran    | saran@gmail.com | 9876543210 | 8.5 | 101 |
| 3 | Rahul    | rahul@gmail.com | 9876543212 | 7.8 | 101 |
| 7 | Sneha    | sneha@gmail.com | 9876543216 | 9.1 | 101 |
| 10 | Harsha   | harsha@gmail.com | 9876543219 | 8.7 | 101 |
+-----+-----+-----+-----+-----+
```

4 rows in set (0.00 sec)

Query OK, 0 rows affected (0.04 sec)

```
mysql> DELIMITER //
```

```
mysql> CREATE PROCEDURE DisplayPlacedStudents()
```

```
-> BEGIN
```

```
-> SELECT
```

```
-> s.student_name, c.company_name, p.package_lpa FROM placements p
```

```
-> JOIN students s
```

```
-> ON p.student_id = s.student_id
```

```
-> JOIN companies c
```

```
-> ON p.company_id = c.company_id;
```

```
-> END //
```

Query OK, 0 rows affected (0.03 sec)

```
mysql> DELIMITER ;
```

```
mysql> CALL DisplayPlacedStudents();
```

```
+-----+-----+-----+
| student_name | company_name | package_lpa |
```

```
+-----+-----+-----+
```

```
| Priya      | Infosys    | 5.2 |
| Anjali     | Infosys    | 5.2 |
| Sneha      | Cognizant   | 5.8 |
| Harsha     | Cognizant   | 5.8 |
| Deepa      | Accenture   | 6.0 |
```

```
+-----+-----+-----+
```

5 rows in set (0.00 sec)

Query OK, 0 rows affected (0.02 sec)

mysql> DELIMITER //

mysql> CREATE PROCEDURE InsertStudent(

-> IN p_name VARCHAR(100),

-> IN p_email VARCHAR(100),

-> IN p_phone BIGINT,

-> IN p_cgpa DECIMAL(3,1),

-> IN p_dept_id INT)

-> BEGIN

-> INSERT INTO students (student_name,email,phone,cgpa,dept_id)

-> VALUES

-> (p_name, p_email, p_phone,p_cgpa, p_dept_id);

-> END //

Query OK, 0 rows affected (0.01 sec)

mysql> DELIMITER ;

mysql> CALL InsertStudent('Ramesh', 'ramesh@gmail.com',9876543220, 8.4,101);

Query OK, 1 row affected (0.02 sec)

```
mysql> SELECT *
```

```
-> FROM students
```

```
-> WHERE email = 'ramesh@gmail.com';
```

```
+-----+-----+-----+-----+-----+
| student_id | student_name | email      | phone   | cgpa | dept_id |
+-----+-----+-----+-----+-----+
|      11 | Ramesh      | ramesh@gmail.com | 9876543220 | 8.4 |    101 |
+-----+-----+-----+-----+-----+
```

```
1 row in set (0.00 sec)
```

```
mysql> DELIMITER //
```

```
mysql> CREATE PROCEDURE UpdateStudentCGPA(
```

```
-> IN p_student_id INT,
```

```
-> IN p_new_cgpa DECIMAL(3,1))
```

```
-> BEGIN
```

```
-> UPDATE students
```

```
-> SET cgpa = p_new_cgpa
```

```
-> WHERE student_id = p_student_id;
```

```
-> END //
```

```
Query OK, 0 rows affected (0.01 sec)
```

```
mysql> DELIMITER ;
```

```
mysql> CALL UpdateStudentCGPA(1, 9.0);
```

```
Query OK, 1 row affected (0.01 sec)
```

```
mysql> SELECT
```

```
-> student_id,
```

```
-> student_name,
```

```
-> cgpa
```

```
-> FROM students
```

```
-> WHERE student_id = 1;
```

```
+-----+-----+-----+
| student_id | student_name | cgpa |
```

```
+-----+-----+-----+
```

```
|      1 | Saran      | 9.0 |
```

```
+-----+-----+-----+
```

1 row in set (0.00 sec)

mysql> DELIMITER //

mysql> CREATE TRIGGER trg_set_application_status

-> BEFORE INSERT ON applications

-> FOR EACH ROW

-> BEGIN

-> SET NEW.status = 'Applied';

-> END //

Query OK, 0 rows affected (0.11 sec)

mysql> DELIMITER ;

mysql> INSERT INTO applications

-> (application_id, student_id, job_id, status)

-> VALUES

-> (11, 1, 2, NULL);

Query OK, 1 row affected (0.05 sec)

mysql> SELECT * FROM applications WHERE application_id = 11;

```
+-----+-----+-----+-----+
```

```
| application_id | student_id | job_id | status |
```

```
+-----+-----+-----+-----+
```

```
|      11 |      1 |    2 | Applied |
```

```
+-----+-----+-----+-----+
```

1 row in set (0.04 sec)


```
mysql> DELIMITER //
```

```
mysql> CREATE TRIGGER trg_check_cgpa
```

```
-> BEFORE INSERT ON students
```

```
-> FOR EACH ROW
```

```
-> BEGIN
```

```
-> IF NEW.cgpa < 6 THEN
```

```
->     SIGNAL SQLSTATE '45000'
```

```
->     SET MESSAGE_TEXT = 'Student CGPA must be 6 or above';
```

```
-> END IF;
```

```
-> END //
```

```
Query OK, 0 rows affected (0.06 sec)
```

```
mysql> DELIMITER ;
```

```
mysql> INSERT INTO students
```

```
-> (student_name, email, phone, cgpa, dept_id)
```

```
-> VALUES
```

```
-> ('Test Student', 'test@gmail.com', 9876543210, 5.5, 101);
```

```
ERROR 1644 (45000): Student CGPA must be 6 or above
```

```
mysql> INSERT INTO students
```

```
-> (student_name, email, phone, cgpa, dept_id)
```

```
-> VALUES
```

```
-> ('Ravi', 'ravi@gmail.com', 9876543210, 7.5, 101);
```

```
Query OK, 1 row affected (0.01 sec)
```

```
mysql> CREATE TABLE students_backup (
```

```
-> student_id INT,
```

```
-> student_name VARCHAR(100),
```

```
-> email VARCHAR(100),
```

```
-> phone BIGINT,
```

```
-> cgpa DECIMAL(3,1),  
-> dept_id INT,  
-> deleted_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP);
```

Query OK, 0 rows affected (0.04 sec)

```
mysql> DELIMITER //
```

```
mysql> CREATE TRIGGER trg_backup_deleted_student
```

```
-> BEFORE DELETE ON students  
  
-> FOR EACH ROW  
  
-> BEGIN  
  
-> INSERT INTO students_backup (  
-> student_id,  
-> student_name,  
-> email,  
-> phone,  
-> cgpa,  
-> dept_id )  
-> VALUES (  
-> OLD.student_id,  
-> OLD.student_name,  
-> OLD.email,  
-> OLD.phone,  
-> OLD.cgpa,  
-> OLD.dept_id );  
  
-> END //
```

Query OK, 0 rows affected (0.05 sec)

```
mysql> DELIMITER ;
```

```
mysql> DELETE FROM students
```

```
-> WHERE student_id = 11;
```

Query OK, 1 row affected (0.05 sec)

mysql> SELECT * FROM students_backup;

student_id	student_name	email	phone	cgpa	dept_id	deleted_at
11	Ramesh	ramesh@gmail.com	9876543220	8.4	101	2026-06-17 10:12:58

1 row in set (0.00 sec)